• **DRIVEABILITY—DIAGNOSTIC TROUBLE CODES (DTC) P1285 AND/OR P1299—ENGINE OVERHEAT CONDITION** • **ELECTRICAL—LAMP—MALFUNCTION INDICATOR LAMP (MIL) ILLUMINATED—DIAGNOSTIC TROUBLE CODES (DTC) P1285 AND/OR P1299—ENGINE OVERHEAT CONDITION** • **COOLING SYSTEM—ENGINE OVERHEAT** • **ENGINE—ENGINE OVERHEAT**	**Article No. 01-21-11**

FORD: 2002 THUNDERBIRD

LINCOLN: 2001-2002 LS

ISSUE

Some vehicles may exhibit an engine overheating condition and/or the Malfunction Indicator Lamp (MIL) may also be illuminated with codes P1285 and P1299 stored in memory. This may be caused by an inoperative Hydraulic Cooling Fan (HCF). This inoperative condition may be intermittent.

ACTION

Replace the HCF motor with revised part XW4Z-8K621-AA to correct the condition. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

1. After the engine has cooled, check the level of the coolant in the degas bottle. It may be necessary to add or remove coolant to/from the degas bottle to bring the level to the cold fill MAX. Tighten the cap until it clicks several times indicating a tight seal. Check for and correct any coolant leaks in system.

NOTE

THE PURPOSE OF THE DEGAS BOTTLE IS TO ALLOW FOR EXPANSION OF THE COOLANT WHEN IT GETS HOT. IF THERE IS NO AIR-SPACE IN THE BOTTLE (OVERFILLED CONDITION), THE HOT COOLANT WILL DISPLACE COOLANT INSTEAD OF AIR, RESULTING IN WHAT APPEARS TO BE AN "OVERHEAT" OR "BOIL OVER" CONDITION.

2. Check for and correct any leaks in the Hydraulic Cooling Fan (HCF) system.
3. Set the parking brake and disconnect the negative battery cable.
4. Remove the engine appearance cover and the upper radiator sight shield.
5. Loosen the band clamp at the throttle body on the air inlet hose.
6. Remove the retaining nut from the air inlet hose (3.9L only).
7. Remove the retaining bolt from the air filter housing.

NOTE

REINSTALL THIS BOLT BACK INTO THE INNER FENDER FOR FUTURE REFERENCE. DO NOT USE THIS BOLT IN THE FAN SHROUD ON REASSEMBLY OR RADIATOR DAMAGE WILL OCCUR.

8. Disconnect electrical connectors at the Mass Airflow Sensor and Air Temperature Sensor.
9. Disconnect all hoses connected to the air inlet tube and note their position.

NOTE

ON 3.0L ENGINES ONLY: DISCONNECT THE THROTTLE POSITION (TP) SENSOR AND THE IDLE AIR CONTROL (IAC) VALVE ELECTRICAL CONNECTORS.

10. Remove the air filter housing and inlet tube as one assembly.
11. Remove the right and left radiator upper support brackets.

Copyright © 2001 Ford Motor Company

12. Remove the engine appearance cover support bracket from the left cylinder head (3.9L only).
13. Remove the two bolts holding the lower radiator hose to the left cylinder head. Slide the hose down and rearward approximately 1″ (3.9L only).
14. Free the HCF motor low pressure hose from the shroud by releasing the two (2) retainers located on the left side and below the pump to allow hose movement.
15. Remove the high-pressure hose bracket bolt at the fan motor.
16. Remove the two (2) fan shroud bolts on either side and lift slightly the fan shroud to clear the lower radiator support.
17. Push the shroud rearward toward the engine to create enough clearance to remove the fan blade. Maintain the opening on the left side for clearance to the fan (use a shim, rubber mallet, block of wood, plastic bottle, etc.).
18. Remove the center fan nut and fan blade.

NOTE
FAN NUT IS REVERSE THREAD.

19. Remove three (3) fan motor bolts, then remove the fan shroud shim device to allow the shroud to move forward.
20. Remove the fan motor from the shroud (hoses still attached) being careful to keep the motor at/above the fluid fill line on the reservoir.
21. Remove the high and low-pressure hoses from the fan motor keeping the motor above the reservoir to avoid leaks.
22. Replace HCF Motor, part number XW4Z-8K621-AA.
23. Replace the Teflon high-pressure hose washer, part number 388898-S.

NOTE
CHECK THE HIGH-PRESSURE HOSE FITTING FOR TEFLON WASHER SHAVINGS THAT COULD CONTAMINATE THE SYSTEM AND COMPROMISE HCF PERFORMANCE.

24. Reverse procedure to reinstall fan motor and engine components.

NOTE
WHEN INSTALLING THE FAN MOTOR INTO THE SHROUD, INSTALL THE HIGH PRESSURE HOSE BRACKET BOLT TO HELP POSITION THE MOTOR IN PLACE. THIS WILL PREVENT THE MOTOR FROM ROCKING AROUND AIDING ASSEMBLY.

NOTE
LIGHTLY OIL THE THREADS OF THE FAN MOTOR ATTACHING BOLTS FOR EASE OF INSTALLATION.

25. Refill the HCF reservoir with MERCON® and restore battery connections.
26. Check and clear all codes from the PCM.

Torque Values:

- Fan Motor to Shroud: 14 N•m (11 lb-ft.)
- Fan Blade to Fan Motor: 25 N•m (18.4 lb-ft.)
- High Pressure Line Bracket retaining Nut: 5 N•m (3.7 lb-ft.)

Refer to Workshop Manual section 303-03 for additional information and torque values.

PART NUMBER	PART NAME
XW4Z-8K621-AA	Motor - Hydraulic Cooling Fan
388898-S	Teflon Washer - High-Pressure Hose

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And California Emissions Warranty Coverage

OPERATION	DESCRIPTION	TIME
012111A	Replace Hydraulic Engine Cooling Fan Motor Lincoln LS 3.0L Engine	1.5 Hrs.
012111B	Replace Hydraulic Engine Cooling Fan Motor Lincoln LS And Thunderbird 3.9L Engine	1.7 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
8K621	42

OASIS CODES: 402000, 499000, 698298